CNuvS Ex. 9 - Jan Lindauer, Paul Feidieker

Task 5

(a) Kendall's queueing notation:

- **A**: Arrival process
- **S**: Service process
- **m**: Number of servers
- **N**: places in the system (bounded queue length), if not given, then assumed inf
- **K**: Population Size
- **SD**: Queue discipline

A and **S** are noted as follows:

- **M**: Exponential process (Markovian)
- **D**: Deterministic
- **G**: General

[1, p. 36]

(b) The notation is structured as A/S/m, where each component gives a specific rule about how the system behaves.

The first **M** in M/M/1 represents **A**, an exponential process, specifically a Markovian.
The second **M** represents **S**, also an exponential process, specifically a Markovian.
The **1** stands for **m**, the number of servers.
[1, p. 36, 37, 44]

(c)

Sources

[1] Prof. Dr. Marco Zimmerling, “CNuvS 05: Queueing Theory.” 2026.